CFD study with FireFOAM and FDS in reduced scale well-ventilated compartment fire scenarios

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Abstract

The paper focuses on large eddy simulations of reduced scale, well-ventilated compartment fire scenarios and aims to analyse and compare CFD results as obtained with the current modelling capabilities of FireFOAM and FDS. To this purpose, a qualitative and quantitative comparison of the results with the two CFD codes as a function of grid size is reported, for a range of fuel flow rates and ventilation flow rates. Both codes correctly predict flame tilting and a relatively uniform smoke layer. Yet, there are obvious differences between the predictions with the model settings as applied in this study. More specifically, FireFOAM predicts better the fire plume dynamics, in terms of maximum flame temperatures, with significantly less grid dependency. On the other hand, FDS predicts slightly better the smoke layer temperatures for all the grid sizes considered. An analysis of the flow field inside the enclosure is presented. The observed deviations between the codes are discussed and qualitatively analysed. Overall, there were no significant differences in the predicted air supply flow rates between the codes, which remained within 15% of the experimental measurements.

Keywords: compartment fires, mechanical ventilation, flame tilting, FireFOAM, FDS, CFD

1. Introduction

Within the framework of compartment fires, the use of Computational Fluid Dynamics (CFD) modelling has, nowadays, become an integral part of fire safety

science and engineering. CFD-based modelling applied to fire research has made significant progress over the past decades [1]. Many of the physical processes that control fire dynamics can now be modelled with suitable accuracy, e.g., the turbulent flow, the pressure dynamics, the combustion process, the thermal radiation transport and soot production.

Overall, there is a need for numerical models that can accurately simulate a plethora of fire-related scenarios (e.g., design of smoke control systems, fire detection, flame spread, fire suppression) over a wide range of time and length scales. A deeper understanding of the capabilities and limitations of the current CFD models is essential for providing the fire safety engineers with the appropriate guidelines, if they want to use CFD to design and assess fire safety systems, such as ventilation systems, in buildings and industrial facilities.

Given their significance in fire safety engineering, compartment fires have been the focus of many CFD studies in the past: giving an overview of the challenges and the modelling capabilities of CFD to simulate such scenarios (e.g., [2, 3, 4]); trying to validate CFD models in compartment fire scenarios (e.g., [5, 6, 7]); applying CFD to study various complex enclosure fire scenarios (e.g., [8, 9]; and discussing various important aspects involved in compartment fires (e.g., [10, 11]). The present study builds upon existing knowledge gained from previous studies and aims to contribute in the field of compartment fire modelling.

The main objective of the paper is to evaluate the performance and predictive capabilities of two of the most widely used CFD codes (i.e., FireFOAM and FDS) when it comes to simulating (using LES) the fire dynamics in well-ventilated compartment scenarios with their default models, for a range of fuel flow rates and mechanical smoke extraction rates (with natural air supply through an inlet branch). Including different grid sizes, where the flames can vary from being fairly well-resolved to under-resolved, the paper examines whether the CFD codes can accurately predict the flame and smoke temperatures (i.e., near the fire source and at the ceiling level) inside the enclosure. As such, no particular emphasis is put on the prediction of (minor) combustion products. In addition, a qualitative comparison and analysis of the flame characteristics predicted by the two CFD codes will be made.

However, there are significant differences between the two CFD codes (e.g., related to their sub-models for turbulence, combustion and radiation, near-wall modelling, heat transfer and discretization schemes, among many others). Furthermore, there can be compensating effects from the application of certain sub-models in each CFD code which are impossible to determine a-priori in the simulations. A more comprehensive comparison between the two CFD codes con-

sidering multiple compartment fire scenarios (as is done in, e.g., [14]) would be required in order to make more concrete statements regarding the capabilities of each code to simulate such scenarios. However, such an extensive task cannot be undertaken in a single scientific paper. As such, it is clear that a direct comparison between FireFOAM and FDS is not straightforward and, hence, is not attempted in the analysis presented in the paper. Instead, the predictive capabilities of each code will be evaluated, to the extent possible, independently from each other and possible reasons for observed discrepancies between simulations and experiments will only be qualitatively discussed.

2. Modelling

Two different CFD codes are considered for the Large Eddy Simulations (LES) presented in the paper: FireFOAM version 1912 [12], developed by FM Global; and the Fire Dynamics Simulator (FDS) version 6.9.1 [13], developed by the National Institute of Standards and Technology (NIST). Both CFD codes have been developed to simulate fire-related scenarios (e.g., involving smoke and heat transport, fire plumes, liquid pool fires, compartment fires, fire extinction, fire suppression, water sprays and flame spread) with FDS having an extensive validation guide [14]. An overview of the main governing equations considered in FireFOAM and FDS can be found in [15] and [16], respectively.

A detailed outline of the main sub-models used in FireFOAM has been previously reported in [17], while for FDS it has been presented in [18]. Hence, only a brief outline is presented here in order for the readers to be aware of some of the major differences in the modelling between the two codes. It should be noted that the intention is to evaluate how each code performs independently and not to have a direct comparison, hence, there was no attempt to use exactly the same sub-models. However, the same geometry and boundary conditions (i.e., considering 1D heat transfer through the enclosure walls and neglecting any leakages) were used in both codes. With FDS, the default models in LES mode were employed [13]. With FireFOAM, the default model settings, previously reported in [19], have been applied.

2.1. Turbulence modelling

- FireFOAM
  Turbulence is modelled with the dynamic Smagorinsky model [20] which calculates the sub-grid scale viscosity, μ_{sgs}, as:

$$\mu_{sgs} = \max(\overline{\rho}(c_s\Delta)^2|\widetilde{S}|, -\mu) \tag{1}$$

where Δ is the filter width (i.e., cube root of the cell volume) and $\widetilde{S}$ is the resolved strain rate. The model also accounts for back-scattering [21], which can occur near the base of weakly-turbulent buoyant fires, by allowing the sgs viscosity to take negative values (i.e., down to $-\mu$), but keeping the effective dynamic viscosity positive (i.e., $\mu_{eff} = \mu + \mu_{sgs} \geq 0$).

The sgs kinetic energy and sgs dissipation rate are estimated as [22]:

$$k_{sgs} = c_I \Delta^2 |\widetilde{S}|^2 \tag{2}$$

$$\epsilon_{sgs} = \frac{c_\epsilon k_{sgs}^{3/2}}{\Delta} \tag{3}$$

The model parameters, c_s, c_I, and c_ϵ are calculated based on a dynamic procedure as local averages of their face values. More details regarding the dynamic procedure employed can be found in [15]. The dynamic procedure for calculating the model parameters should, in theory, be better in capturing the quasi-laminar to turbulent transition of the flow, compared to models using constant model parameters, typically encountered in scenarios involving buoyancy-driven flows (see [15] for results with FireFOAM). Nevertheless, the performance of both types of turbulence models (i.e., involving constant and dynamic coefficients), with respect to grid size, still needs to be further investigated.

- FDS

  Turbulence is modelled using a modified Deardorff model [23] which estimates the sub-grid scale viscosity, μ_{sgs}, as:

$$\mu_{sgs} = \overline{\rho} C_v \Delta \sqrt{k_{sgs}} \tag{4}$$

  where $C_v = 0.1$ [24] is a model constant and Δ is the filter width (i.e., cube root of the cell volume).

  The sub-grid scale kinetic energy, k_{sgs}, is calculated from an algebraic relationship, based on scale similarity, as:

$$k_{sgs} = \frac{1}{2}((\overline{u} - \hat{\overline{u}})^2 + (\overline{v} - \hat{\overline{v}})^2 + (\overline{w} - \hat{\overline{w}})^2) \tag{5}$$

  where $\overline{u}$ is the LES filtered velocity at length scale Δ and $\hat{\overline{u}}$ is a weighted average of u over the adjacent cells (i.e., test-filtered at length scale 2Δ).

The terms $\overline{\nu}$, $\hat{\overline{\nu}}$, $\overline{w}$, $\hat{\overline{w}}$ are defined similarly [16]. It is worth noting that the considered turbulence model has been successfully applied with FDS in a wide range of fire scenarios in the past [14].

It is worth noting that the range of Reynolds numbers involved in typical fire scenarios, particularly above fire sources, are much lower than the ones encountered in highly turbulent flows. As such, differences in turbulence modelling can be one of the influential factors in the resulting fire plume structure, particularly when coarser grid sizes are employed (see [15]).

2.2. Combustion modelling

The combustion model considers a one-step, infinitely fast, irreversible chemical reaction between fuel and oxidizer. Turbulence-chemistry interactions are modelled using the eddy dissipation model (EDM) [25] in FireFOAM while a turbulent batch reactor model (which reduces to the Eddy Dissipation Concept (EDC) as described in [26] in this case) is employed in FDS. As such, the fuel reaction rate in both models is calculated as:

$$\overline{\dot{\omega}_F'''} = -\overline{\rho}\frac{\min(\widetilde{Y}_F, \widetilde{Y}_{O_2}/s)}{\tau_{mix}} \tag{6}$$

where $\widetilde{Y}_F$ and $\widetilde{Y}_{O_2}$ are the resolved fuel and oxygen mass fractions, respectively and s is the stoichiometric oxygen-to-fuel mass ratio.

- FireFOAM
  The mixing time scale model considers both laminar and turbulent conditions, which are assumed to be governed by diffusion and sgs advection, respectively. A synthetic turbulent mixing time scale based on the geometric mean of the sgs velocity stretching time scale (Δ/u'), associated with large-scale mixing, and the Kolmogorov time scale ($(\nu/\epsilon_{sgs})^{1/2}$), associated with the mixing on molecular level, is considered.

  The mixing time scale, τ_{mix}, is then calculated as [27]:

$$\tau_{mix} = \min\left(\underbrace{\frac{\Delta^2}{\alpha}}_{\tau_{lam}}, \underbrace{\sqrt{\frac{\Delta}{u'}\left(\frac{\nu}{\epsilon_{sgs}}\right)^{1/2}}}_{\tau_{turb}} \right) \tag{7}$$

  where α is the thermal diffusivity and $u' = (2k_{sgs}/3)^{1/2}$ is the rms of the sgs turbulent velocity fluctuations. Validation of this mixing time scale model has been previously reported in [19] for a variety of fire plume scenarios.

- FDS
  The mixing time scale, τ_{mix}, is calculated by comparing the mixing times for diffusion (i.e., τ_d), sub-grid scale advection (i.e., τ_u), and buoyant acceleration (i.e., τ_g) as [28]:

$$\tau_{mix} = \max(\tau_{chem}, \min(\tau_d, \tau_u, \tau_g, \tau_{flame})) \tag{8}$$

  with

$$\tau_d = \frac{\Delta^2}{D_F}, \quad \tau_u = \frac{C_u \Delta}{\sqrt{(2/3)k_{sgs}}}, \quad \tau_g = \sqrt{\frac{2\Delta}{g}} \tag{9}$$

  where D_F is the effective diffusivity of the fuel, and $C_u = 0.4$ [14] is a model constant. The buoyant time scale is based on a constant acceleration and is expected to control mixing when using coarse grids. The mixing time scale, τ_{mix}, is also bounded between a low (i.e., τ_{chem}) and a big value (i.e., τ_{flame}). This mixing time scale model has been extensively validated for a variety of fire scenarios in the past (see [14]).

It is worth noting that both these two mixing sub-models have been evaluated with the same solver (i.e., FireFOAM) in the past (see [27]) and both performed equally well, as a function of grid size, in a variety of fire plume scenarios in open atmosphere. As such, it is not believed that differences in combustion modelling could be a major reason for potential differences observed in the predictions between the two CFD codes in the present study. Nevertheless, no compartment fire scenarios were considered in [27] and, as such, the performance of the mixing sub-model in FireFOAM needs to be further evaluated in such scenarios.

2.3. Radiation modelling

The radiation intensity is a function of spatial location and angular direction and is obtained by solving the radiative transfer equation (RTE), for a non-scattering medium, based on the finite volume method. Both gas and soot contribute to the emission and absorption of thermal radiation, while a simple soot yield model (i.e., 0.024 for C_3H_8 [29]) is used here to account for soot in both CFD codes.

- FireFOAM
  Absorption and emission is modelled with the weighted-sum-of-gray-gases

(WSGG) model [30] with the total absorption coefficient, including the contribution from soot, calculated as:

$$\kappa = \underbrace{-\frac{\ln(1-\epsilon)}{L}}_{\text{gas}} + \underbrace{C_{soot} f_v T}_{\text{soot}} \tag{10}$$

where $f_v = (\overline{\rho}\widetilde{Y}_s)/\rho_s$, $C_{soot} = 1862$ m^{-1}K^{-1} [31] and $\rho_s = 1800$ kg/m^3 [32]. The total emissivity, ϵ, of the mixture is calculated as the sum of four fictitious gray gases while the path length is estimated as $L = 3.6V/A$.

The divergence of the radiative heat fluxes is then estimated as:

$$\nabla \cdot \overline{\dot{q}_r''} = \underbrace{4\sigma\kappa\widetilde{T}^4 \underbrace{\left(1 + 6C_{TRI}\frac{\widetilde{T''^2}}{\widetilde{T}^2}\right)}_{\text{(TRI)}}}_{\text{emission}} + \underbrace{E\overline{\dot{q}'''}}_{\text{correction}} - \underbrace{\kappa G}_{\text{absorption}} \tag{11}$$

where σ is the Stefan-Boltzmann constant, where the TRI term is calculated according to [33] with the model constant $C_{TRI} = 2$ [34], $\widetilde{T''^2}$ is the temperature variance, $\overline{\dot{q}'''}$ is the heat release rate per unit volume, and G is the total irradiance.

The correction, E, in the emission term is calculated as:

$$E = \max\left(\chi_{r,exp} - \underbrace{\frac{\iiint_V (\text{emission} - \text{absorption}) dV}{\iiint_V \overline{\dot{q}'''} dV}}_{\chi_{r,\text{simulated}}}, 0\right) \tag{12}$$

which ensures the desired amount of heat being emitted locally by radiation (i.e., equal to $\chi_{r,exp}$) in case the flames are under-resolved (i.e., coarse-grid simulations). When flames are sufficiently resolved, the local radiative fraction is an outcome of the CFD simulations.

- FDS

  The total absorption coefficient, κ, of the gas mixture is calculated as the sum of the individual gases' gray coefficients, κ_i, as:

$$\kappa = \sum_i \kappa_i(\widetilde{Y}_i, \widetilde{T}) \tag{13}$$

In turn, $\kappa_{n,i}$ are calculated by a narrow-band model, RADCAL [35], considering the minimum of the Planck mean coefficient (dependent on the local composition and temperature) and the effective absorption coefficient (dependent on the local composition, temperature and path length L) as:

$$\kappa_i = \min(\kappa_{i,P}, \kappa_{i,e}) \tag{14}$$

The default value for the path length in FDS is $L = 0.1$ m. For more details regarding the calculation of the two absorption coefficients the reader is referred to [16].

In regions where combustion takes place (i.e., $\chi_R \overline{\dot{q}'''}$ ¿ 1 kW/m^3), the emission source term can be under-predicted if coarse grids are used. As such, a correction term, C, is applied which is calculated as:

$$C = \frac{\sum(\chi_r \overline{\dot{q}'''} + \kappa G)V}{\sum(4\kappa\sigma\widetilde{T}^4)V} \tag{15}$$

where $\overline{\dot{q}'''}$, is the heat release rate per unit volume,χ_r is the radiative fraction of the fuel and V is the cell volume. In addition, the correction term is bounded between 0.1 and 100 to prevent spurious behaviour at the start of the simulations.

Finally, the radiative heat loss term in the energy equation are estimated as:

$$\nabla \cdot \overline{\dot{q}''_r} = \kappa(4\sigma C\widetilde{T}^4 - G) \tag{16}$$

where G is the total irradiance.

Based on the above, it is clear that there are differences in how the radiative correction term is evaluated between the two codes. In FireFOAM, χ_r, is fixed in every cell by correcting the emission term if it is under-predicted (i.e., fixing the global radiative emission) while in FDS the prescribed χ_r value defines the minimum amount of global radiative emission. As such, it is not guaranteed that radiative emission will be the same in the two codes (i.e., in terms of the global predicted χ_r value). With the current approach, the prescribed χ_r value in FireFOAM ideally needs to account for potential radiative emission from the smoke layer, which is not easy to evaluate a-priori in the simulations and which can negatively influence the predicted temperature field. These aspects can have a direct impact on the predicted temperatures inside in the flame region, but also in the

smoke layer temperatures far away from the fire plume. The potential impact of the different way of evaluating the path length between the two codes, which can also be linked back to the predicted (flame / smoke layer) temperatures, is not straightforward to assess, unless a detailed sensitivity study is conducted (which was not part of the objectives of the paper).

2.4. Other modelling options

It should be noted that there are also (significant) differences in near-wall modelling, heat transfer modelling and discretization schemes, among many other aspects, between the two codes. Also, by default, an extinction model based on an enthalpy balance [16] is employed in FDS in LES mode (as also considered in this paper) while no extinction model was considered with FireFOAM. Even though well-ventilated experimental scenarios are considered in this study it is unclear whether some (minor) local effect due to extinction modelling will be present in the CFD predictions with the lowest extraction flow rates. There will naturally also be differences in the solution algorithm and implementation differences of the sub-models between the two codes. It is beyond the scope of the paper to list in detail all the differences between the two codes.

3. Experimental set-up

Experiments performed by ASNR (formerly IRSN) in the reduced-scale enclosure NYX [36] are considered for validation purposes of the simulations. A schematic of the experimental setup, along with the position of the instrumentation, is presented in Figure 1. To avoid uncertainties associated with under-ventilated conditions and flame extinction modelling, only well-ventilated scenarios are considered in this work. The internal dimensions of the enclosure are (L x W x H) 1.5 m x 1.25 m x 1 m. The floor, side walls and ceiling consist of a 23 mm thick layer of calcium silicate ($\rho = 770$ kg/m^3, $k = 0.185$ W/m/K, $c_p = 970$ J/kg/K). A square C_3H_8 burner (i.e., 18 cm x 18 cm) with three fuel flow rates, $\dot{m}_F$, (i.e., 0.1 g/s, 0.2 g/s, 0.3 g/s), is used as fire source. The burner has a height of 15 cm and it is positioned in the center of the enclosure.

The radiative fractions, χ_r, for the 0.1 g/s, 0.2 g/s and 0.3 g/s fire sources, measured in free-burn conditions, are 0.17, 0.21 and 0.23, respectively. Both the air supply and the extraction lines used for ventilation are positioned at the top of the enclosure. The supply duct is open, allowing for fresh air to enter the enclosure, while the extraction line is connected to a fan, equipped with a variable speed drive, with a fixed extraction rate (i.e., initial value before the start of the

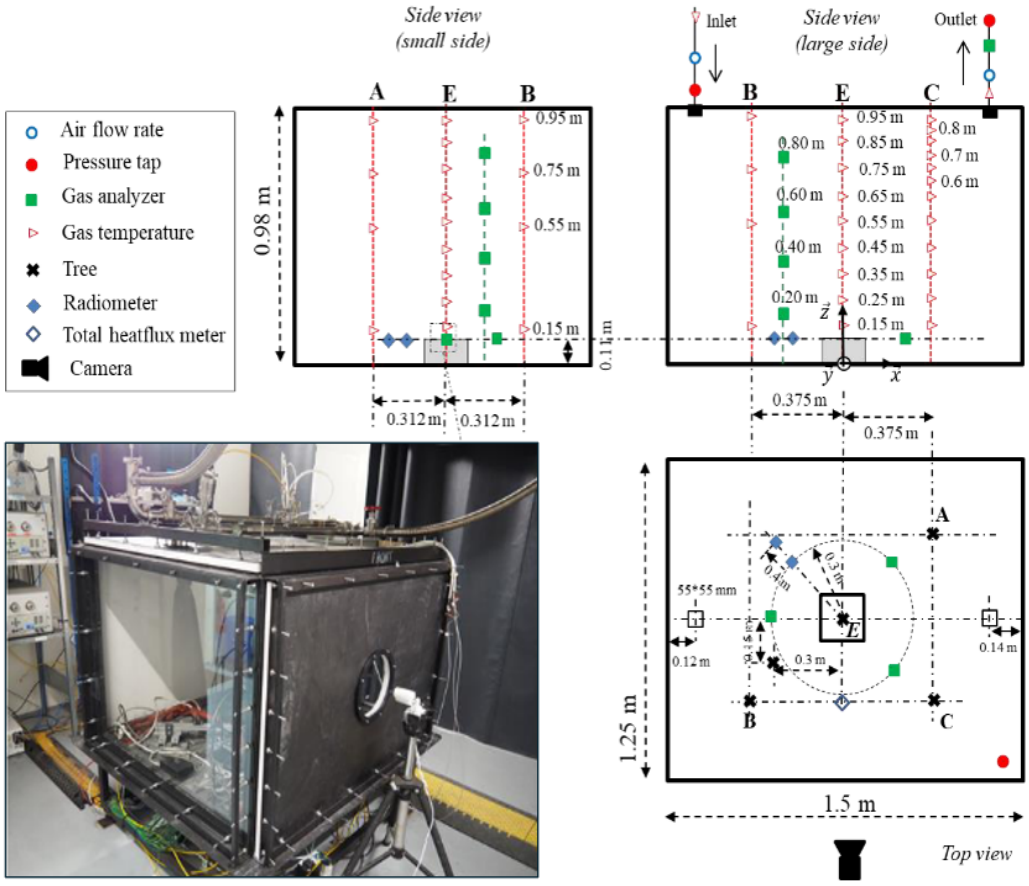


Figure 1: Schematic of the NYX facility and the instrumentation position in the experiments.

fire). The volumetric extraction rates gradually increase after the burner is turned on and during the tests, due to the generation of hot combustion products of lower density, until they eventually reach steady-state values after a given time. The duration of each experimental test was 1300 s. An overview of the different experimental tests considered (i.e., fuel flow rates as well as initial and steady-state extraction flow rates) in the simulations is presented in Table 1. The gas temperatures above the fire and at other locations inside the enclosure are measured by 0.5 mm K type thermocouple trees. To get a better feel of the experimental setup, Figure 2 presents instantaneous images from a selection of scenarios with varying fuel mass flow rate and a fixed extraction flow rate (i.e., $\dot{V}_{extr.}^{ini.}$ = 50 m^3/h) to qualitatively illustrate flame tilting that occurred towards the air supply pipe. For more

Table 1: Overview of the experimental conditions considered in the simulations. More information about the experiments performed at ASNR (formerly IRSN) can be found in [36].

Fuel flow rate - $\dot{m}_F$ (g/s)	0.1	0.1	0.2	0.2	0.3	0.3
Extraction flow rate (initial) - $\dot{V}_{extr.}^{ini.}$ (m^3/h)	20	50	30	50	40	50
Extraction flow rate (steady-state) - $\dot{V}_{extr.}^{steady}$ (m^3/h)	26	60	44	68	61	76

detailed information regarding the experimental facility and the tests performed the readers are referred to [36].

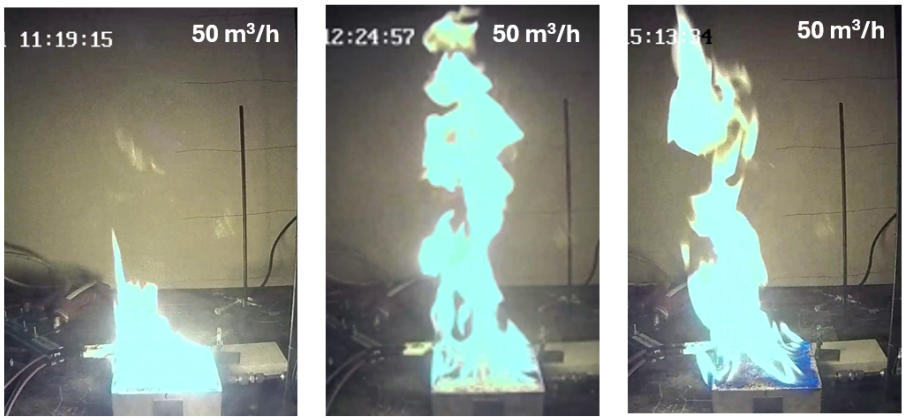


Figure 2: Instantaneous plots from a selection of experiments with different fuel mass flow rates and $\dot{V}_{extr.}^{ini.}$ = 50 m^3/h illustrating flame tilting towards the air supply side.

4. Numerical set-up

A computational domain with dimensions (L x W x H) of 1.52 m $\times$ 1.26 m $\times$ 1 m is employed to model the compartment in the considered scenarios. The supply and extraction lines are modelled as 0.3 m long square pipes of dimensions 6 cm x 6 cm. The thickness and properties of the compartment walls (made out of calcium silicate) have been set according to the experiments and 1D heat transfer is considered. The ambient temperature and pressure are set to 20°C and 101325 Pa, respectively. The radiative fraction values for the different fuel flow rates of propane have been set to the values reported experimentally. Leakages are not considered in the simulations.

The classical way to determine the required grid size for unconfined flames a priori is the use of the D^* criterion, which examines the ratio of D^*/Δ. Ratio values exceeding 10 have been reported to be sufficient in the past with FDS [18]. This is also within the range of reported values that has proven to guarantee sufficient accuracy in simulations of various fire scenarios in the past [14]. Table 2 shows that almost all scenarios considered in the simulations fulfil the D^* criterion requirements. It is worth noting that the computational cost associated with uniform grid sizes finer than 1 cm was too prohibitive to be considered in the current study. Given the type of compartment scenario, with important flow in the entire enclosure, the use of a local grid refinement strategy was not deemed suitable. Therefore, the three (uniform) grid sizes considered are 0.03 m, 0.02 m, and 0.01 m, leading to total numbers of cells in the gas phase of 73 k, 251 k, 2 mil., respectively.

However, there are additional requirements imposed on the mesh resolution by the geometry of interest. It must be noted that the size of the supply / extraction ducts is limited (i.e., only 6 cm x 6 cm cross-section). This not only poses a limit to the coarsest mesh size that could be employed in the simulations, but it also implies that even with the finest mesh, there are only 6 cells across the side of the ducts (and only 2 or 3 on the other meshes). This may limit the accuracy of the flows inside the ducts, as well as of the flow of the incoming air jet, as discussed below. Hence, not much attention will be given to these results (i.e., they are included mainly as part of a grid sensitivity study). Nevertheless, these grid resolutions are still used here as well in order to test the predictive capabilities of the two CFD codes on coarser grids, which could be used in practice for faster engineering-type of simulations. Hence, it is considered relevant to demonstrate the significant impact the mesh size has on the results here.

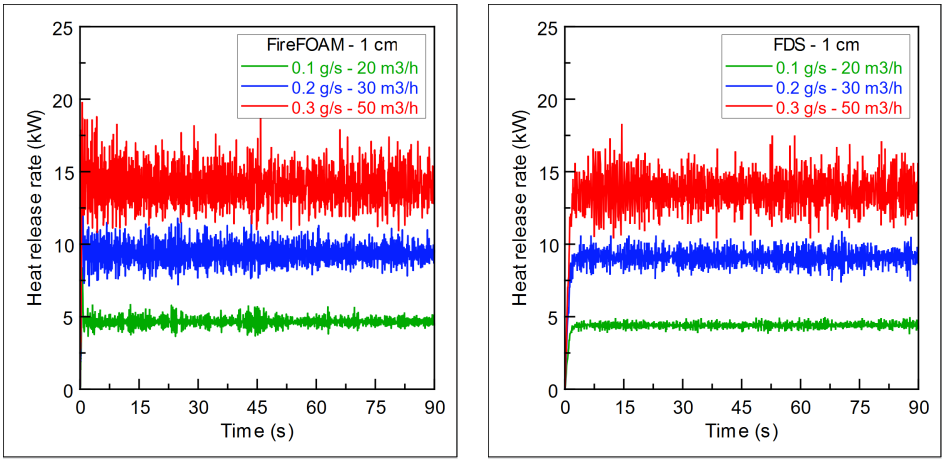
As 1D heat transfer is considered in the solid structures, there are also cells in the solid phase (i.e., walls and ceiling) which are mapped from the gas phase cells. Through the thickness of the walls, 50 cells are used in the solid phase. As such, the total number of cells in the solid phase (i.e., for the compartment walls) were 420 k, 960 k, 3.84 mil for the 3 cm, 2 cm and 1 cm grid size in the gas phase, respectively. The simulations were run for 90 s and averaged over the last 60 s, hence allowing for 30 s for any initial transient effects to leave the computational domain. To illustrate this, an overview of the predicted heat release rate (HRR) inside the compartment with FireFOAM and FDS for different scenarios and for the finest grid size in the gas phase (i.e., 1 cm) is presented in Figure 3. A more quantitative comparison of the predicted HRRs with both codes is presented later (see Table 5). Note, however, that the considered running time is not sufficient to

Table 2: Overview of the grid resolutions, in terms of D^*/Δ, for the different scenarios.

$\dot{m}_F$ (g/s)	D^{*a} (m)	$D^*/0.03$ (-)	$D^*/0.02$ (-)	$D^*/0.01$ (-)
0.1	0.11	3.74	5.61	11.22
0.2	0.15	4.94	7.40	14.81
0.3	0.17	5.80	8.71	17.41

[a]Characteristic fire diameter: $D^* = \left(\frac{\dot{Q}}{\rho_\infty c_p T_\infty \sqrt{g}}\right)^{2/5}$ where $\dot{Q}$ (kW) is the heat release rate of the fire, ρ_∞ (kg/m^3), c_p (kJ/(kg·K)), T_∞ (K) are the density, specific heat and temperature of the ambient air, respectively, while g (m/s^2) is the gravitational acceleration.

accurately capture the heat transfer through the compartment walls as the duration of the experiments was substantially longer (i.e., up to 1300 s).



(a) (b)

Figure 3: Predicted HRR for a selection of scenarios with (a) FireFOAM and (b) and FDS with a grid size of 1 cm in the simulations.

5. Results

5.1. Mesh sensitivity

Before discussing the simulation results in detail, a brief qualitative impression of the mesh sensitivity is given through the average gas temperature and

vertical velocity on 2D slices across the center of the burner for two scenarios (i.e., the lowest and highest fuel and extraction rates considered), as presented in Figures 4 and 5, respectively. Note that the effect of grid size is also included in the quantitative analysis presented later in the paper.

There is a clear effect of the grid resolution in the flame temperature predictions for both codes in the lowest fuel and extraction rate scenario considered (i.e., $\dot{m}_F$ = 0.1 g/s and $\dot{V}_{extr.}^{ini.}$ = 20 m^3/h). In this case, the flame region only starts being well-resolved with the finest grid resolution (i.e., 1 cm) with distinct differences observed between FireFOAM and FDS in the flame lengths and resulting flame structures. The effect of grid size diminishes for the highest fuel and extraction rate scenario considered (i.e., $\dot{m}_F$ = 0.3 g/s and $\dot{V}_{extr.}^{ini.}$ = 50 m^3/h), which is to be expected considering the D^* criterion (i.e., increasing characteristic fire diameter with higher HRR resulting in better resolution in the simulations). Here, FireFOAM's predictions are less grid-sensitive than those of FDS. The effect of dynamic turbulence modelling is believed to be an important reason for the grid insensitivity observed in the predictions with FireFOAM. The model parameter, c_s (Eq.(1)), varied significantly along the centerline above the burner: from about 0 - 0.01 (just above the burner) to 0.13 - 0.16 (just below the ceiling), depending on the case and the grid size considered. Due to its dynamic formulation, the model parameter essentially self-adjusts depending on the local flow conditions (i.e., low values just above the burner, where quasi-laminar conditions are present, to higher values further downstream, where the flow becomes more turbulent due to buoyancy). Qualitatively, some differences in the resulting vertical velocities are also present between the different cases. These can be naturally linked to the differences in the predicted flame temperatures (i.e., higher flame temperatures result in higher axial velocities). To better quantify the differences between the two codes, Table 3 presents a comparison of the predicted maximum flame and thermocouple temperatures, axial velocities and their corresponding locations, for the scenarios depicted in Figures 4 - 5.

5.2. Qualitative analysis

A qualitative overview of the simulation results as obtained with FireFOAM and FDS, illustrating the average gas temperature and vertical velocity on 2D slices across the center of the burner with the finest grid size (i.e., 1 cm) for a selection of scenarios (i.e., fixing the extraction rate and varying the fuel flow rate), is presented first in Figures 6 and 7, respectively. To better quantify the differences between the two codes, Table 4 presents a comparison of the predicted maximum

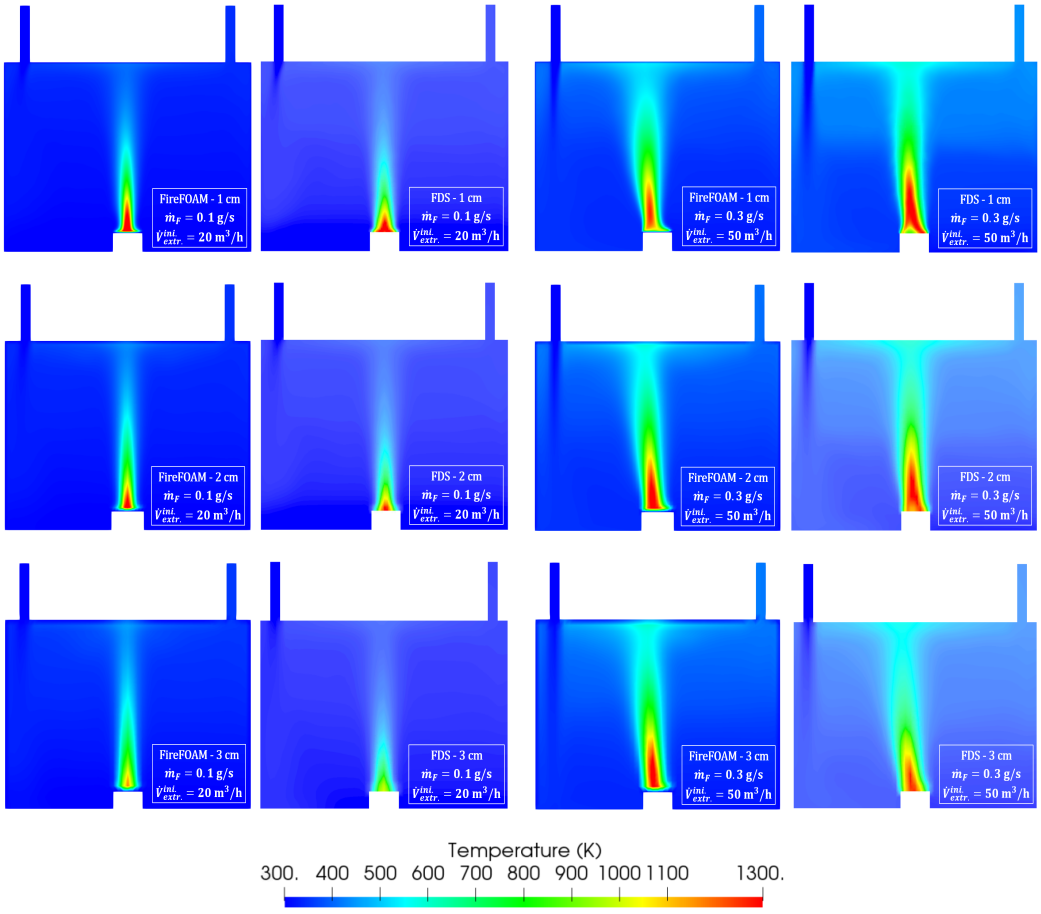


Figure 4: 2D slice of average gas temperature with FireFOAM (first and third column) and FDS (second and fourth column) for $\dot{m}_F$ = 0.1 g/s and $\dot{V}^{ini.}_{extr.}$ = 20 m^3/h (first two columns) and for $\dot{m}_F$ =0.3 g/s and $\dot{V}^{ini.}_{extr.}$ = 50 m^3/h (last two columns). Grid sensitivity study: 3 cm (bottom row), 2 cm (middle row) and 1 cm (top row).

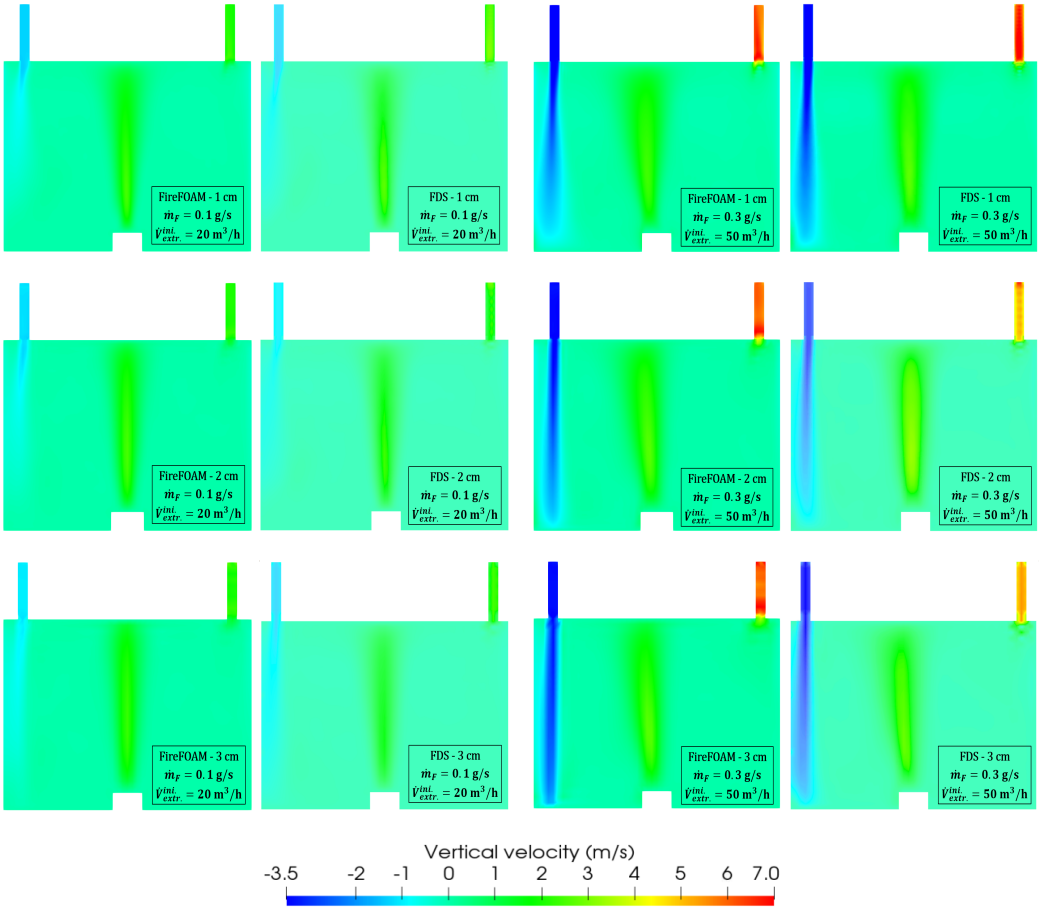


Figure 5: 2D slice of average vertical velocity with FireFOAM (first and third column) and FDS (second and fourth column) for $\dot{m}_F$ = 0.1 g/s and $\dot{V}^{ini.}_{extr.}$ = 20 m^3/h (first two columns) and for $\dot{m}_F$ = 0.3 g/s and $\dot{V}^{ini.}_{extr.}$ = 50 m^3/h (last two columns). Grid sensitivity study: 3 cm (bottom row), 2 cm (middle row) and 1 cm (top row).

Table 3: Overview of the predicted maximum flame temperatures, T_{max}, thermocouple temperatures, TC_{max}, and axial velocities, v_{max}, alongside their corresponding axial locations, $z_{@T,TC_{max}}$ and v_{max}, at thermocouple tree E for the scenarios depicted in Figures 4 - 5.

Case	T_{max} (K)	TC_{max} (K)	$z_{@T,TC_{max}}$ (m)	v_{max} (m/s)	$z_{@v_{max}}$ (m)
$\dot{m}_F = 0.1$ g/s & $\dot{V}^{ini.}_{extr.} = 20$ m^3/h					
FireFOAM - 1 cm	1423	1246 (+12%)	0.04	2.43	0.21
FireFOAM - 2 cm	1413	1222 (+10%)	0.02	2.49	0.3
FireFOAM - 3 cm	1170	1054 (-5%)	0.03	2.34	0.36
FDS - 1 cm	1419	1241 (+12%)	0.03	2.14	0.25
FDS - 2 cm	1411	1229 (+11%)	0.0	2.17	0.20
FDS - 3 cm	968	913 (-18%)	0.0	1.56	0.27
Experiment	-	1109	0.05	-	-
$\dot{m}_F = 0.3$ g/s & $\dot{V}^{ini.}_{extr.} = 50$ m^3/h					
FireFOAM - 1 cm	1048	994 (0%)	0.02	1.50	0.53
FireFOAM - 2 cm	1291	1161 (+17%)	0.02	1.65	0.26
FireFOAM - 3 cm	1244	1121 (+13%)	0.03	1.72	0.27
FDS - 1 cm	1341	1208 (+22%)	0.06	2.36	0.36
FDS - 2 cm	1349	1216 (+23%)	0.12	2.74	0.28
FDS - 3 cm	1305	1166 (+18%)	0.0	1.29	0.51
Experiment	-	990	0.0	-	-

flame and thermocouple temperatures, axial velocities and their corresponding locations, for the scenarios depicted in Figures 6 and 7.

In general, the predicted maximum temperatures in the flame region (i.e., near the burner surface) appear to be consistently lower with FireFOAM, compared to FDS, particularly for the lowest fuel flow rates (see Table 4). The use of a dynamic turbulence model in FireFOAM could be one of the reasons that makes the fire plumes more turbulent near their base and lowers the maximum flame temperatures. Indeed, lower c_s values have been reported in previous work with FireFOAM near the base of pool fire scenarios (see [15]), implying less turbulent mixing and dissipation due to lower μ_{sgs} values (compared to the same turbulence model using the theoretical fixed c_s constant). Whereas this is in line with the observations for mean temperature, the multiple other difference sub-models in each CFD code also have an impact on temperature, making it impossible to state firmly that turbulence modelling is the main reason for the differences observed between the two codes.

Differences in the local radiative flame emission in this region (i.e., related to the correction applied in the radiative emission term) could also affect the resulting flame temperatures. Other differences related to heat transfer modelling e.g., heating of the walls and near-wall modelling, among others, could also play a lo-

cal role in the developed flow field and the mixing occurring inside the enclosure (i.e., particularly near the burner). Nevertheless, compensating effects from the use of different sub-models in each code cannot be excluded. A detailed sensitivity study on a variety of models and their parameters would be required to further pinpoint the exact reasons for the observed differences between the codes (which was not part of the objectives of this study).

Flame tilting towards the air supply pipe, due to the incoming fresh air on this side of the enclosure and the flames looking for fresh oxygen, is predicted by both codes, but the effect is more pronounced with FireFOAM, particularly for the smaller fuel flow rates. The effect of dynamic turbulence modelling, potentially making the fire plumes near their base more turbulent and not allowing for as strong buoyancy forces to develop, and differences in near-wall modelling are two plausible reasons for the stronger tilting observed with FireFOAM. It is worth noting that a similar flame tilting has been observed experimentally after qualitative analysis of the video footage (see Figure 2), which confirms the trends predicted by the simulations. This tilting effect leads to differences in the predicted flame temperatures above the burner presented later in the paper. In the absence of a more quantitative analysis of flame tilting from the experiments, the predicted gas temperatures above the burner will serve as an indicator as to how well tilting is predicted in the simulations.

Overall, consistently higher temperatures (by up to 9% for the highest fuel flow rate case) are predicted with FDS inside the enclosure, regardless of the fuel flow rate considered in the simulations (see Figure 6). This effect also contributes to differences in the predicted radiative fractions between the two codes (presented later in the paper). This is potentially linked to differences in radiation modelling between the two codes (i.e., in the calculation of the absorption coefficients, in the correction of the radiative emission and the value of path length used). The typically longer flame lengths observed with FDS for all fuel flow rates, with the flow hitting the ceiling and being partially re-circulated inside the enclosure towards the floor, could also make the extraction of any hot gases more difficult (i.e., leading to increased temperatures inside the enclosure).

The resulting flow field inside the compartment (see Figure 7), illustrated by the vertical velocities predicted by both CFD codes, reveals similar patterns. There is a strong jet of fresh air flowing into the compartment, down to the floor, while high velocities are predicted inside the extraction line due to low density hot combustion products being removed from the compartment. In addition, an increase in the vertical velocity is observed above the burner due to the buoyant flow generated by the burner. In this region, some minor differences between the

two codes are evident, which can be linked to differences in the predicted flame temperatures as well. Again, the lack of sufficient turbulence at the base of the fire plumes (i.e., use of a turbulence model with constant parameters), can result in higher dissipation of the flow and the development of more laminar-like structures with higher flame temperatures and resulting axial velocities.

Table 4: Overview of the predicted maximum flame temperatures, T_{max}, thermocouple temperatures, TC_{max}, and axial velocities, v_{max}, alongside their corresponding axial locations, $z_{@T,TC_{max}}$ and v_{max}, at thermocouple tree E for the scenarios depicted in Figures 6 and 7.

Case	T_{max} (K)	TC_{max} (K)	$z_{@T,TC_{max}}$ (m)	v_{max} (m/s)	$z_{@v_{max}}$ (m)
$\dot{m}_F = 0.1$ g/s & $\dot{V}_{extr.}^{ini.} = 50$ m^3/h					
FireFOAM - 1 cm	1254	1138 (+26%)	0.01	0.57	0.08
FDS - 1 cm	1470	1272 (+40%)	0.01	1.58	0.24
Experiment	-	906	0.0	-	-
$\dot{m}_F = 0.2$ g/s & $\dot{V}_{extr.}^{ini.} = 50$ m^3/h					
FireFOAM - 1 cm	1171	1081 (+13%)	0.01	1.15	0.51
FDS - 1 cm	1394	1234 (+29%)	0.04	2.18	0.24
Experiment	-	955	0.0	-	-
$\dot{m}_F = 0.3$ g/s & $\dot{V}_{extr.}^{ini.} = 50$ m^3/h					
FireFOAM - 1 cm	1048	994 (+0%)	0.02	1.50	0.53
FDS - 1 cm	1341	1209 (+22%)	0.08	2.36	0.36
Experiment	-	990	0.0	-	-

An overview of the averaged horizontal velocity at height $z = 0.05$ m (i.e., 5 cm above the burner surface) for various fuel flow rates and $\dot{V}_{extr.}^{ini.} = 50$ m^3/h with both codes and for the finest grid size scenarios (i.e., 1 cm) is presented Figure 8. The flow field, determined by a combination of the mechanical extraction and the fire-induced buoyancy, is complex and determines the mixing of air and fuel (and hence, affects the flame tilting). As the jet of incoming fresh air from the air supply pipe hits the compartment floor it is deflected and spreads radially inside the enclosure. Eventually, a big part of this incoming air participates in the combustion above the burner while a small part can eventually reach the extraction pipe where it is taken out of the enclosure. The observed flame tilting, previously seen in Figure 6 with FireFOAM, is in line with the non-uniform horizontal velocity obtained at this axial location (i.e., higher negative horizontal velocity on the right side than positive horizontal velocity on the left side of the burner). This observation is consistent for all fuel flow rates examined. On the other hand, this effect is mainly seen with FDS on the higher fuel flow rates (i.e., 0.1 g/s and 0.2 g/s), as it is also confirmed by the horizontal velocity distribution

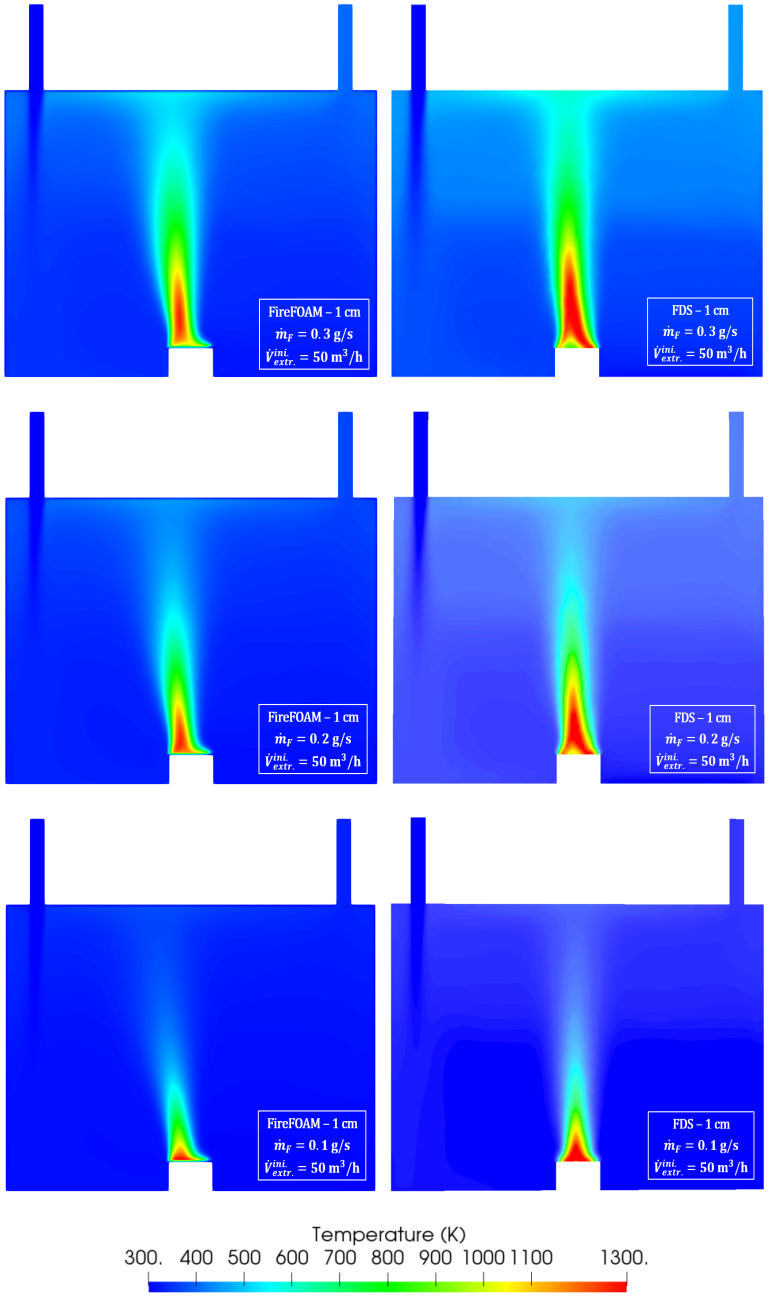


Figure 6: 2D slice of averaged gas temperature with FireFOAM (left) and FDS (right) for $\dot{m}_F$ equal to 0.1 g/s (bottom), 0.2 g/s (middle) and 0.3 g/s (top) with $\dot{V}^{ini.}_{extr.} = 50$ m^3/h and a grid size of 1 cm.

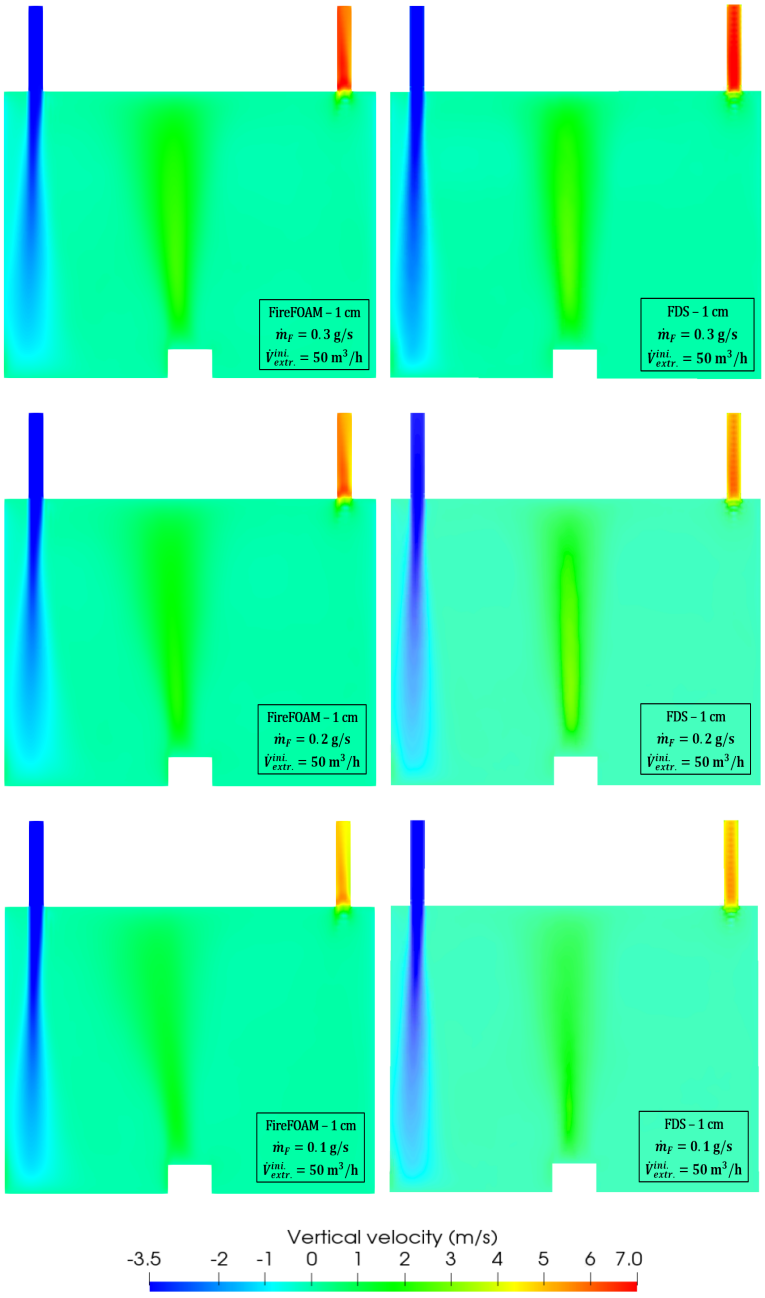


Figure 7: 2D slice of averaged vertical velocity with FireFOAM (left) and FDS (right) for $\dot{m}_F$ equal to 0.1 g/s (bottom), 0.2 g/s (middle) and 0.3 g/s (top) with $\dot{V}^{ini.}_{extr.} = 50$ m^3/h and a grid size of 1 cm.

just above the burner.

5.3. Quantitative analysis

Before discussing graphs, Table 5 provides information on a number of integral quantities obtained from the numerical simulations with both codes. There are some minor quantitative differences, up to 5 - 6$\%$ depending on the case, in the predicted heat release rates between the two codes with the values of FireFOAM being higher. The values of FireFOAM are also slightly higher than the experiments, indicating there was no 100% combustion efficiency in the experiments, as can be expected. In general, the predicted HRR values of FDS are in excellent agreement with the experiments. A reason for the differences between the two codes can be the slightly different values used by default for the heat of combustion for propane (i.e., 46447 kJ/kg in FireFOAM versus 45497 kJ/kg in FDS), as well as the the presence of an extinction model by default in FDS which could potentially have a minor effect (even if the scenarios are in theory well-ventilated).

It is worth noting that the same volumetric extraction flow rate is applied for both codes and that the air supply line is open for fresh air to be entrained inside the enclosure. The predicted air supply flow rates (m^3/h) for the different scenarios are overall, given the uncertainties, within a reasonable degree of accuracy from the experimental values (i.e., deviations on the order of $\pm 15\%$), with diminishing discrepancies as the fuel flow rate increases. A larger fire requires a higher extraction rate (i..e, resulting in higher flow velocities) for well-ventilated-conditions to be maintained inside the compartment. As such, the flow field inside the compartment in the case of a stronger fire is mainly determined by the flow field induced by the extraction rate itself, and less influenced by the flame dynamics. Hence, discrepancies in the predicted flame temperatures (hence, in the resulting flow field) can lead to errors in the predicted air supply rates easier for smaller fires compared to bigger fires. Eventually, the enclosure reaches a slightly different steady-state condition with each code, which is dependent on the predicted fire plume dynamics and the ventilation characteristics at the supply and extraction lines. Nevertheless, the differences in the predicted entrance flow rates between the two CFD codes are, in most cases, less than 5%.

A quantitative comparison between the predictions with FireFOAM and FDS is illustrated in Figures 9 - 11, presenting the predicted temperatures at thermocouple tree E (i.e., indicative of the flame temperature above the burner) and thermocouple trees A-B (i.e., indicative of the temperature variation along the enclosure height). The readers are referred to Fig. 1 for the exact position of the thermocouple trees inside the enclosure. A 15% uncertainty (error bars) has been considered

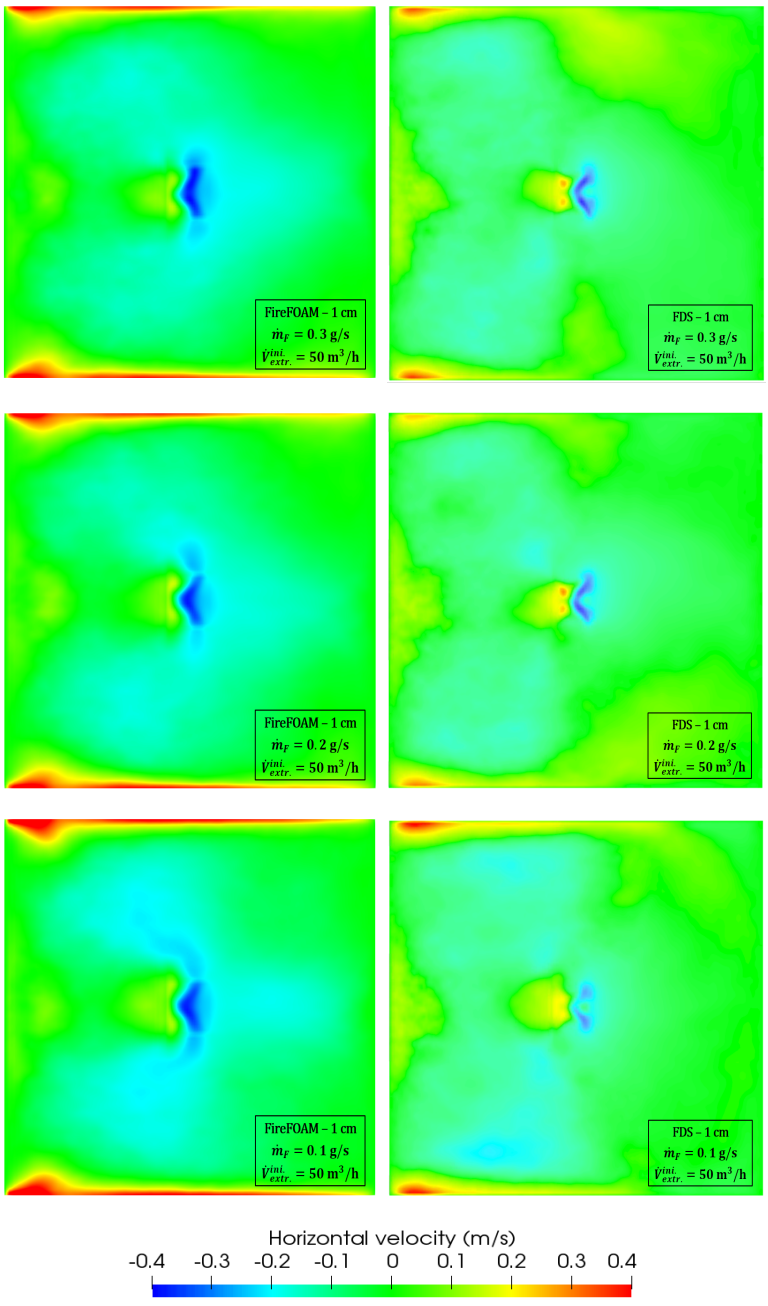


Figure 8: 2D slice of averaged horizontal velocity with FireFOAM (left) and FDS (right)at height $z = 0.05$ m (i.e., 5 cm above the burner surface) for $\dot{m}_F$ equal to 0.1 g/s (bottom), 0.2 g/s (middle) and 0.3 g/s (top) with $\dot{V}^{ini.}_{extr.} = 50$ m^3/h and a grid size of 1 cm.

Table 5: Comparison of predicted heat release rates (kW), radiative fractions (-) and air supply flow rates (m^3/h) with a grid size of 1 cm in the simulations.

Fuel flow rate - $\dot{m}_F$ (g/s)	0.1	0.1	0.2	0.2	0.3	0.3
Extraction flow rate - $\dot{V}_{extr.}^{ini.}$ (m^3/h)	20	50	30	50	40	50
Heat release rate (kW)						
FireFOAM	4.70	4.71	9.38	9.40	14.04	14.05
FDS	4.42	4.55	9.10	9.10	13.65	13.65
Experiment[a,b]	4.31	4.42	9.07	9.13	13.62	14.17
Radiative fraction (-)						
FireFOAM	0.174	0.168	0.215	0.209	0.239	0.234
FDS	0.272	0.248	0.358	0.332	0.408	0.390
Air supply flow rate (m^3/s)						
FireFOAM	16.8	47.4	25.7	45.7	33.7	45.1
	(-11%)	(-6%)	(-4%)	(-4%)	(-4%)	(0%)
FDS	16.0	46.7	26.6	43.4	35.1	45.6
	(-15%)	(-7%)	(0%)	(-9%)	(0%)	(+2%)
Experiment[b]	18.9	50.3	26.7	47.8	35.1	44.9

[a]Determined based on the Carbon Dioxide Generation (CDG) method.
[b]The experimental values are determined between 1000 - 1200 s in the tests.

in the experimental measurements. The readers are referred to Table 6 for a more detailed quantitative analysis of the results presented in Figure 9.

Overall, the temperatures above the burner (thermocouple tree E), in terms of maximum thermocouple temperatures and subsequent temperature decay, are better predicted with FireFOAM on the finest grid size for the majority of the scenarios, and they exhibit a reasonable grid dependency (see Table 6). The only obvious outlier is the case with $\dot{m}_F = 0.3$ g/s and $\dot{V}_{extr.}^{ini.} = 40$ m^3/h for which the comparison of both codes against the experimental data is poor (i.e., deviations in the flame temperatures by up to 600 K). Based on the experimental results and observations, this scenario was on the edge of being under-ventilated, so that partial flame extinction, not accurately captured in the modelling, could partially explain the differences (i.e., the O_2 was approximately 15% by vol. near the burner based on the experimental measurements). The fact that the extinction model in FDS is active by default though, as opposed to FireFOAM, which did not employ one, should have resulted in better predictions with the former. This, however, is not the case.

On the other hand, the FDS predictions reach grid independency for grid sizes of 1 - 2 cm, but the results differ by up to 200 - 300 K, depending on the case, when compared against the FDS predictions on the coarsest grid size (3 cm). In this case, the former predict higher temperatures in the flame region, compared to FireFOAM and the experiments, along the height of the enclosure (in line with what was previously observed qualitatively in Figure 6). These differences are partially linked to unresolved turbulence close to the burner, which could potentially affect the plume structure (i.e., quasi-laminar flame with high temperature) and result in stronger buoyancy forces upwards. This was explored in FDS with the use of the dynamic Smagorinsky model instead of the default Deardorff model (results not shown here for brevity). In this case, lower temperatures by up to 100 K were predicted in the flame region and flame tilting was slightly more pronounced in certain scenarios (i.e., of low fuel flow rates and high extraction flow rates).

It is also believed that near-wall modelling could also potentially play a role in the differences observed in the flow field between the two codes (e.g., locally affecting the flow field in the vicinity of the burner). This was previously reported in [37] for the same configuration at hand and revealed an influence of the tilting behavior of the fire plumes with FireFOAM and the use of finite rate chemistry depending on the boundary condition used for velocity at the walls. A brief sensitivity study (not shown here for brevity) using 'slip' and 'no-slip' for velocity at the walls was conducted using a 1 cm grid size. Overall, comparable flame

tilting was obtained in this case but some effect in the size of the flame region was observed indicating the the flow field near the burner surface could play a role in the local mixing and resulting flame temperatures.

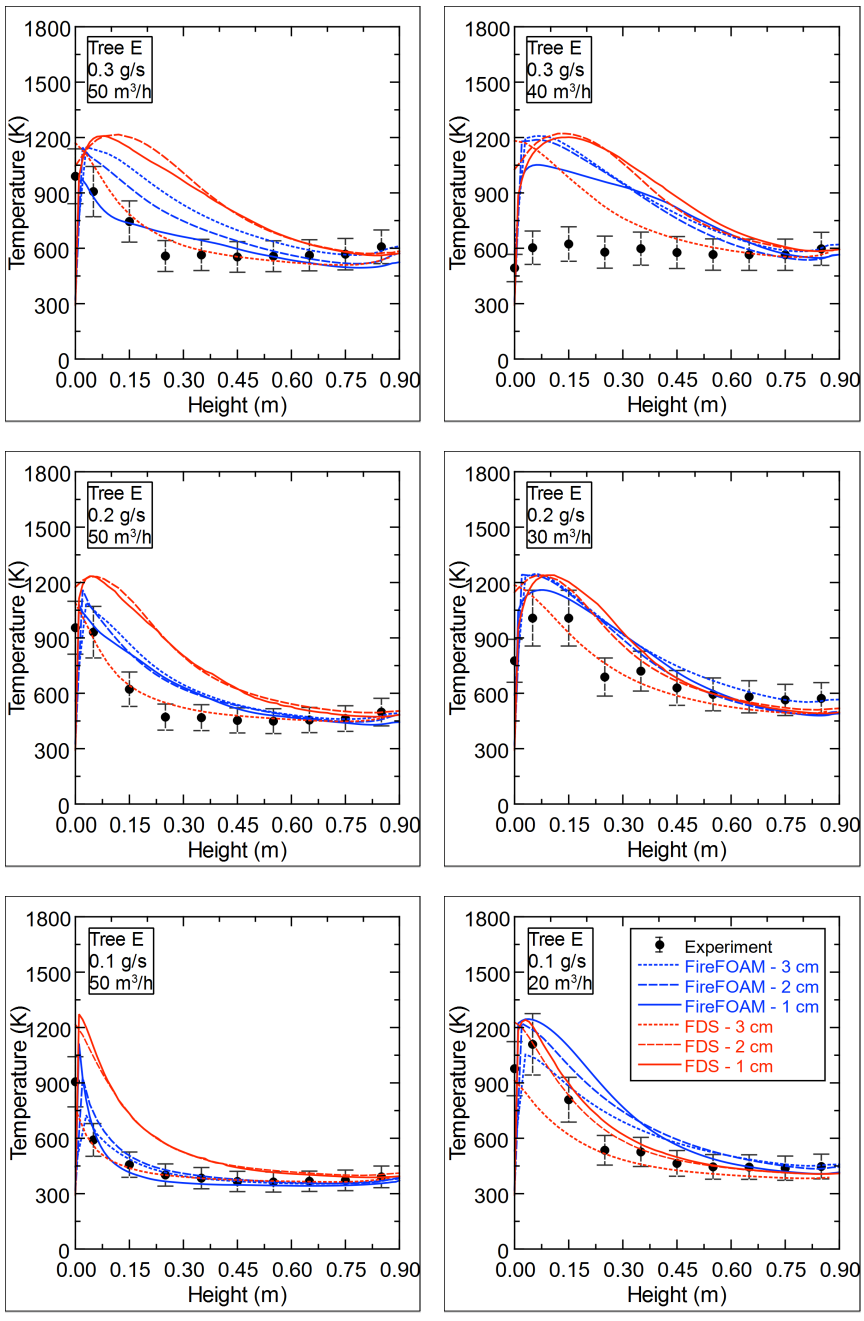
Apart from differences in the boundary condition for velocity, the way sgs quantities are estimated near walls could potentially play a role. An influence, locally in the flow field, near the admission and extraction ducts from not considering the whole ventilation network as in the experiments could also be present. This would require a more detailed and systematic sensitivity study to be conducted which is considered to be beyond the current objectives of this work. In addition, there are no available experimental data of the flow field for direct validation of the CFD results.

A comparison of the predicted temperatures along the height of the enclosure at thermocouple trees A and B is illustrated in Figures 10 - 11. Overall, the results are in line with what was qualitatively previously shown in Figure 6. The FDS predictions, in terms smoke layer temperatures, are consistently somewhat higher than those of FireFOAM (i.e., by up to almost 10%, depending on the case) and are, in most cases, in good agreement with the experiments (i.e., within the considered experimental uncertainty of 15%). The smoke layer temperature predictions between thermocouples trees A and B are also similar, indicating that a fairly uniform smoke layer is correctly predicted by both codes. In general, the discrepancies of the predictions with both codes increase with the fuel flow rate, resulting in maximum deviations of less than 50 K for the cases with a fuel mass flow rate of 0.3 g/s.

6. Conclusions

The study evaluated the predictive capabilities of FireFOAM and FDS in simulating compartment fire scenarios using large eddy simulations and their default model settings to date. To this purpose, a series of experiments performed by ASNR (former IRSN) on a reduced-scale, well-ventilated enclosure with mechanical extraction, were considered. In total, six different scenarios with varying fuel flow rates and extraction flow rates were considered for validation purposes of the simulations.

Overall, the performance of FireFOAM and FDS was satisfactory when compared to the experiments. More specifically, both codes qualitatively predicted the experimentally observed flame tilting, which was in line with non-uniform horizontal velocities obtained above the burner surface, as well as fairly uniform temperatures inside the smoke layer. FireFOAM's predictions of the fire plume

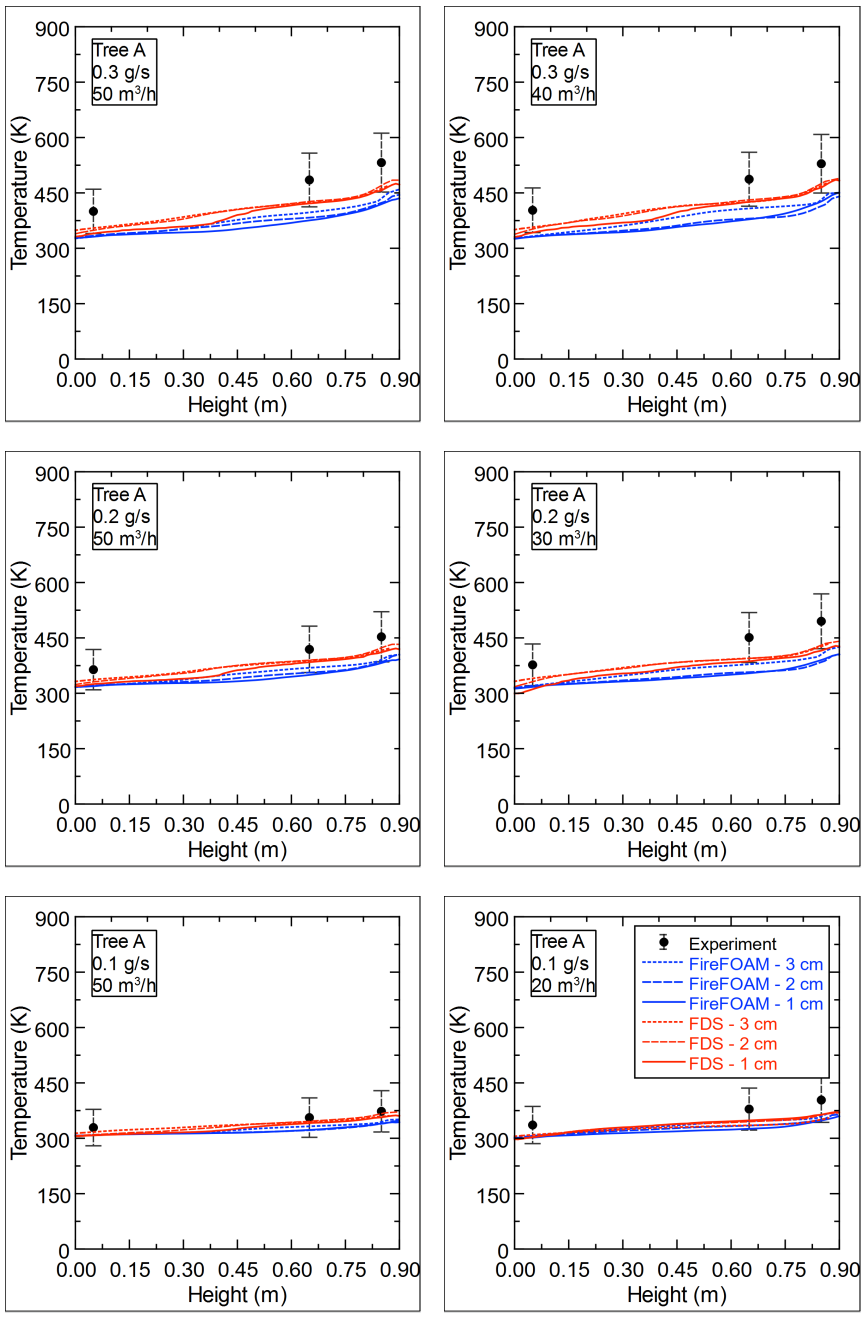


(a) (b)

Figure 9: Average temperatures with FireFOAM and FDS at thermocouple tree E.

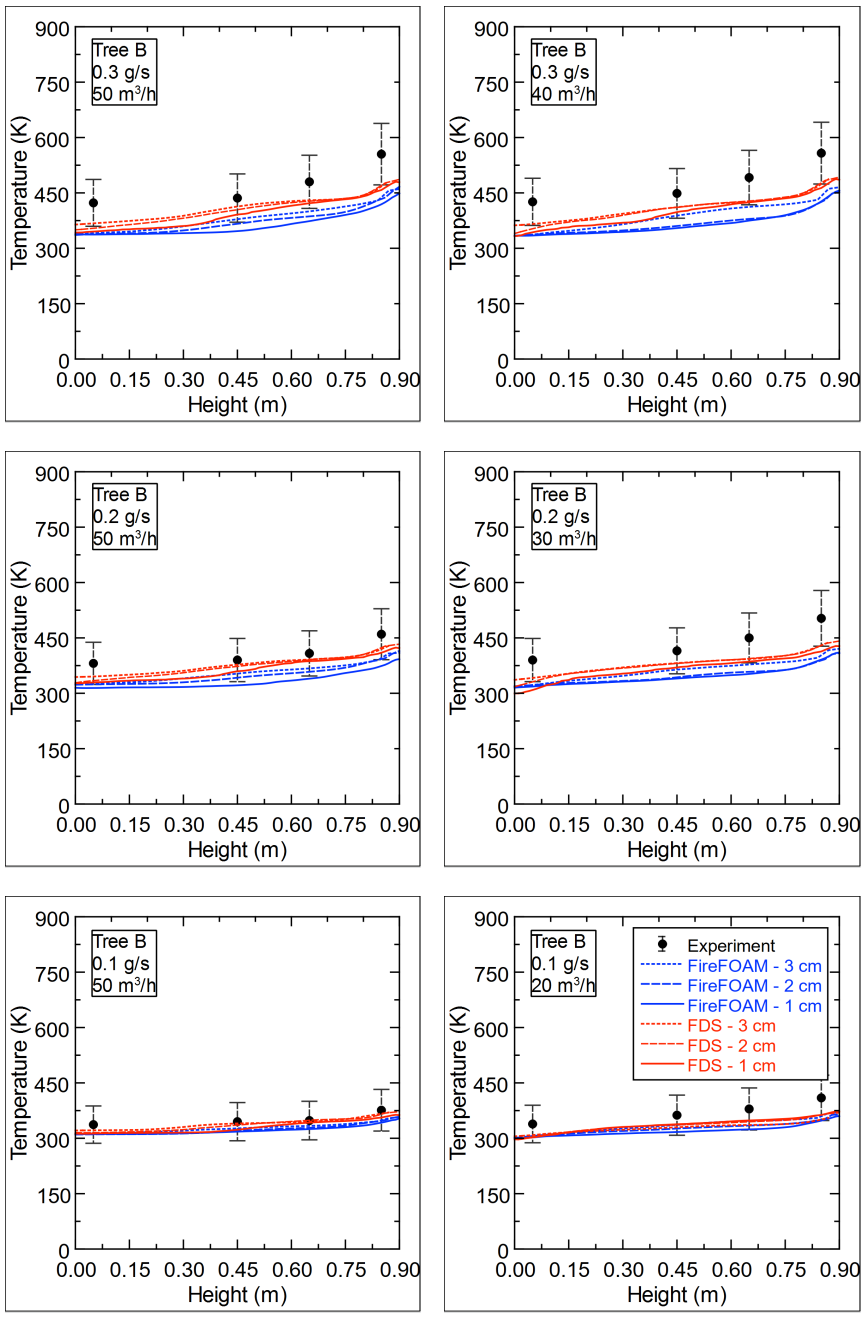
Table 6: Overview of the predicted maximum thermocouple temperatures, TC_{max}, alongside their locations, $z_{@TC_{max}}$ for the scenarios depicted in Figure 9.

Case	TC_{max} (K)	$z_{@TC_{max}}$ (m)
$\dot{m}_F = 0.1$ g/s & $\dot{V}_{extr.}^{ini.} = 20$ m³/h		
FireFOAM (1 cm / 2 cm / 3 cm)	1246 / 1222 / 1054 (+12% / +10% / -5%)	0.04 / 0.02 / 0.03
FDS (1 cm / 2 cm / 3 cm)	1241 / 1229 / 913 (+12% / +11% / -18%)	0.03 / 0.0 / 0.0
Experiment	1109	0.05
$\dot{m}_F = 0.1$ g/s & $\dot{V}_{extr.}^{ini.} = 50$ m³/h		
FireFOAM (1 cm / 2 cm / 3 cm)	1138 / 929 / 731 (+26% / +3% / -19%)	0.01 / 0.02 / 0.03
FDS (1 cm / 2 cm / 3 cm)	1272 / 1217 / 757 (+40% / +34% / -16%)	0.01 / 0.0 / 0.0
Experiment	906	0.0
$\dot{m}_F = 0.2$ g/s & $\dot{V}_{extr.}^{ini.} = 30$ m³/h		
FireFOAM (1 cm / 2 cm / 3 cm)	1159 / 1243 / 1248 (+15% / +23% / +24%)	0.08 / 0.02 / 0.06
FDS (1 cm / 2 cm / 3 cm)	1240 / 1239 / 1186 (+23% / +23% / +18%)	0.11 / 0.08 / 0.0
Experiment	1007	0.05 - 0.15
$\dot{m}_F = 0.2$ g/s & $\dot{V}_{extr.}^{ini.} = 50$ m³/h		
FireFOAM (1 cm / 2 cm / 3 cm)	1081 / 1171 / 1066 (+13% / +23% / +12%)	0.01 / 0.02 / 0.03
FDS (1 cm / 2 cm / 3 cm)	1234 / 1232 / 1089 (+29% / +29% / +14%)	0.04 / 0.06 / 0.0
Experiment	955	0.0
$\dot{m}_F = 0.3$ g/s & $\dot{V}_{extr.}^{ini.} = 40$ m³/h		
FireFOAM (1 cm / 2 cm / 3 cm)	1052 / 1188 / 1209 (+69% / +91% / +94%)	0.06 / 0.06 / 0.06
FDS (1 cm / 2 cm / 3 cm)	1202 / 1221 / 1182 (+93% / +96% / +90%)	0.15 / 0.12 / 0.0
Experiment	623	0.15
$\dot{m}_F = 0.3$ g/s & $\dot{V}_{extr.}^{ini.} = 50$ m³/h		
FireFOAM (1 cm / 2 cm / 3 cm)	994 / 1161 / 1122 (0% / +17% / +13%)	0.02 / 0.02 / 0.03
FDS (1 cm / 2 cm / 3 cm)	1209 / 1216 / 1166 (+22% / +23% / +18%)	0.08 / 0.12 / 0.0
Experiment	990	0.0



(a) (b)

Figure 10: Average temperatures with FireFOAM and FDS at thermocouple tree A.



(a) (b)

Figure 11: Average temperatures with FireFOAM and FDS at thermocouple tree B.

dynamics (i.e., in terms of maximum flame temperatures and subsequent temperature decay) were, in most cases, closer to the experimental measurements. On the other hand, FDS predicted better the smoke layer temperatures across different grid sizes (i.e., within less than 15% of the experimental data). These differences also resulted in higher predicted radiative fractions with FDS (i.e., 50 - 70% depending on the case), a consequence of the different modelling options employed in each code. The resulting air supply flow rates were comparable for both codes (i.e., within $\pm 15\%$ of the experimental values).

The CFD study illustrated the important coupling between fire plume dynamics inside the compartment and the ventilation induced flow field when a mechanical ventilation system is in place. Direct validation of the flow field by means of experimental data would be an important step forward, as it is deemed impossible to isolate the impact of the modelling options as employed in each CFD code. A more comprehensive comparison between the two CFD codes, considering additional compartment fire scenarios, would be required in the future in order to further evaluate the predictive capabilities of each code to simulate such scenarios.

Acknowledgments

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